named it Titan.* It was a dot of light a billion miles away, gleaming in reflected sunlight. From the time of its discovery, when European men wore long curly wigs, to World War II, when American men cut their hair down to stubble, almost nothing more was discovered about Titan except the fact it had a curious, tawny color. Ground-based telescopes could, even in principle, barely make out some enigmatic detail. The Spanish astronomer J. Comas Solá reported at the turn of the twentieth century some faint and indirect evidence of an atmosphere.

In a way, I grew up with Titan. I did my doctoral dissertation at the University of Chicago under the guidance of Gerard P. Kuiper, the astronomer who made the definitive discovery that Titan has an atmosphere. Kuiper was Dutch and in a direct line of intellectual descent from Christianus Huygens. In 1944, while making a spectroscopic examination of Titan, Kuiper was astonished to find the characteristic spectral features of the gas methane. When he pointed the telescope at Titan, there was the signature of methane.* When he pointed it away, not a hint of methane. But moons were not supposed to hold onto sizable atmospheres, and the Earth's Moon certainly doesn't. Titan could retain an atmosphere, Kuiper realized, even though its gravity was less than Earth's, because its upper atmosphere is very cold. The molecules simply aren't moving fast enough for significant numbers to achieve escape velocity and trickle away to space.

Daniel Harris, a student of Kuiper's, showed definitively that Titan is red. Maybe we were looking at a rusty surface, like that of Mars. If you wanted to learn more about Titan, you could also measure the polarization of sunlight reflected off it. Ordinary sunlight is unpolarized. Joseph Veverka, now a fellow faculty member at Cornell University, was my graduate student at Harvard University, and therefore, so to speak, a grandstudent of Kuiper's. In *his* doctoral work, around 1970, he measured the polarization of Titan and found that it changed as the relative positions of Titan, the Sun, and the Earth changed. But the change was very different from that exhibited by, say, the Moon. Veverka concluded that the character of this variation was consistent with extensive clouds or haze on